COM3110/4115: Text Processing

Text Compression: dictionary methods + further topics

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Overview

- Introduction to Text Compression
  - Lossless vs lossy compression
  - Dictionary vs Symbolwise methods
  - Static vs Semi-static vs Adaptive methods
- Symbolwise methods
  - Introduction to Symbolwise methods
  - Coding
    - Huffman Coding
    - Arithmetic Coding
  - Further topics in Symbolwise methods
- Dictionary Methods
  - LZ77 and gzip
- Further topics:
  - Synchronisation
  - Performance Issues

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  - ◇ LZ77 and gzip
- Further topics:
  - ◇ Synchronisation
  - ◇ Performance Issues

This lecture

Dictionary Methods

- Dictionary methods rely on replacing substrings in a text with codewords, or indices, that identify the substring in a *dictionary* or *codebook*
- Compression obtained by codewords being shorter than strings they replace
- Contrast with symbolwise methods – compression obtained by using variable length codes such that frequent symbols get shorter codes

Dictionary Methods (ctd)

- Simple example: *digram coding* — simple method for ASCII text
  - ◇ uses an 8-bit code dictionary where
    - 128 ASCII characters represent themselves
    - 128 codes represent common letter pairs
  - ◇ at worst each 7-bit character is expanded to 8-bits
  - ◇ at best character pairs (14 bits) are reduced to 8 bits
- Such an approach can be extended to include larger dictionary entries

  e.g. common words such as *and* and *the*

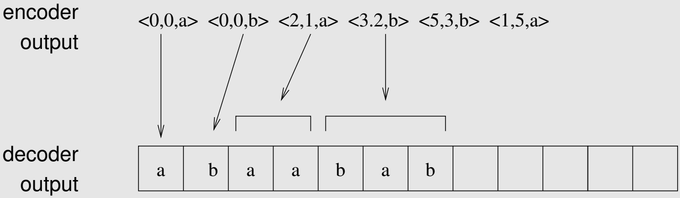
  e.g. common prefixes/suffixes, such as *pre*, *tion*

Dictionary Methods (ctd)

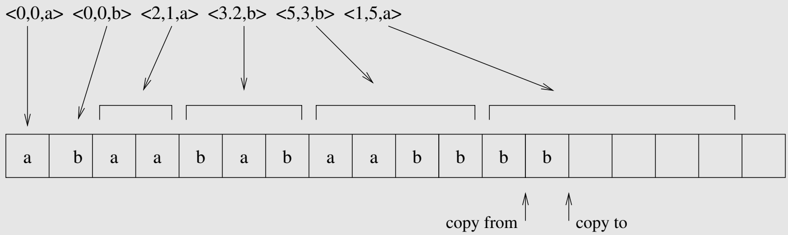
- However, difficult to get good compression with a general dictionary
  - words common to many texts tend to be short
  - dictionary suited to one sort of text, poor for another
- Can move to a semi-static scheme where a new codebook is constructed for each text. *But* some obvious drawbacks:
  - inefficiencies in transmitting dictionary
  - hard to decide which words to select for dictionary
- Solution: use an adaptive dictionary scheme
  - most such schemes based on two methods proposed by Ziv and Lempel in 1977/78 (known as LZ77 and LZ78)
  - key idea: replace substring with a pointer to previous occurrence in same text

Dictionary Methods: LZ77

- LZ77 can be explained most easily in terms of an example of its decoding (example from Witten *et al.*)
- Suppose alphabet is just $\{a, b\}$



- Encoder output is a series of triples
  - ◇ first component indicates how far back in decoded output to look for next phrase
  - ◇ second indicates the length of that phrase
  - ◇ third is next character from input (only necessary when not found in previous text, but included for simplicity)



- ◇ Next, decode characters represented by triple `<5,3,b>`
  - go back 5 characters (to 3rd from start)
  - copy 3 characters (*aab*)
  - add the 3rd item from triple – *b* – to this
- ◇ Next triple, `<1,5,a>`, is a 'recursive reference'
  - go back one character (*b*)
  - *sequentially* copy the next 5 characters

    — when 2nd char needed, it's available (as a copy of first)
  - results in the addition of 5 consecutive *b*'s and an *a*

Dictionary Methods: LZ77 (ctd)

- Various issues must be addressed in implementing an adaptive dictionary method such as LZ77, including . . .
- How far back in the text to allow pointers to refer
  - ◇ references further back increase chance of longer matching strings, but also increase number of bits required to store pointer
  - ◇ typical distance is a few thousand characters
- How large the strings referred to can be
  - ◇ again, the larger the string, the larger the width parameter specifying it
  - ◇ typical value $\sim$ 16 characters
- During encoding, how to search window of prior text for longest match with the upcoming phrase
  - ◇ linear search very inefficient
  - ◇ best to index prior text with a suitable data structure, such as a trie, hash, or binary search tree

Dictionary Methods: *gzip*

- A popular high performance implementation of LZ77 is *gzip*
- Uses a hash table to locate previous occurrences of strings
  - hash accessed by next 3 characters
  - holds pointers to prior locations of the 3 characters
- Pointers and phrase lengths stored using variable length Huffman codes
  - computed semi-statically by processing 64K blocks of data at a time
  - this much can easily be held in memory, so appears as if single-pass
- Pointer triples are reduced to pairs, by eliminating 3rd element
  - common Huffman coding used to transmit *both* phrase lengths *and* characters
    - this is transmitted first
  - if a phrase length was transmitted, then next unit will be a pointer

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Synchronisation

- Good compression techniques work best on large files
  - ◇ decompression techniques inherently sequential
  - ◇ tends to preclude random access
  - ◇ full-text retrieval systems *require random access*
    - need to consider special measures to facilitate random access for these applications
- Good compression methods make random access difficult, because
  - ◇ use variable length codes
    - can't start decoding at random point, since may not be on codeword boundary
  - ◇ use adaptive models
    - cannot determine model without decoding all prior text
- No good solution for adaptive modelling
- Best to use static models for full text retrieval

Synchronisation (ctd)

- Techniques have been developed for achieving random access in compressed files
- **Synchronisation points**
  - assume smallest unit of random access in compressed archive is the *document*
  - *either* store bit offset of document
  - *or* ensure it ends on byte boundary
- **Self-synchronising codes**
  - design code so that regardless of where decoding starts, comes into synchronisation rapidly and stays there
  - problematic for full text retrieval
    - not possible to guarantee how quickly synchronisation will be achieved

Performance Issues

- Performance considerations for compression algorithms include
  - speed
  - memory
  - compression rate (% remaining or % removed)
- Some methods to compare for performance (from Witten *et al.*):
  - tests were performed on a benchmark corpus that included: a novel, fax bitmap, C source code, Excel spreadsheet, executable code, etc.

Method	Description
pack	zero-order, character-based, semi-static Huffman coder
char	zero-order, character-based, adaptive arithmetic coder
ppm	variable-order, character-based, adaptive arithmetic coder
huffword	zero-order, word-based, semi-static Huffman coder
gzip-f	LZ77-type, semi-static coder, fast option
gzip-b	LZ77-type, semi-static coder, best compression
compress	LZ78-type, Unix *compress* utility
null	copy input to output (Unix *cat*)

Performance Issues (ctd)

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Method	Relative Speed		Compression	
	Encoding	Decoding	bpc	% Remaining
pack	0.6	0.9	4.53	56.6
char	2.9	4.0	4.49	56.1
ppm	5.3	5.9	2.11	26.4
huffword	2.2	0.9	2.95	36.9
gzip-f	1.1	0.4	2.91	36.4
gzip-b	7.0	0.3	2.53	31.6
compress	1.0	0.6	3.31	41.4
null	0.2	0.2	8.00	100.0

Performance Issues: gzip

- gzip permits 9 degrees of adjustment for speed vs compression:
  - ◇ 1 = fastest, 9 = best compression
- Figures for Linux gzip compressing Moby Dick on a Macbook Pro (2015), Quad-Core Intel Core i7:

Setting	Time (secs)	File Size (bytes)	Compression (% remaining)
1 (fast)	0.01	581453	47.7
2	0.02	558783	45.8
3	0.03	537084	44.0
4	0.03	524983	43.0
5	0.05	507946	41.6
6	0.07	500234	41.0
7	0.08	498976	40.9
8	0.11	498291	40.8
9 (best)	0.10	498291	40.8
null	0.00	1220150	100.0

Summary

- Symbolise compression methods rely on variable length coding to assign shorter codes to high probability symbols and longer codes to low probability symbols.
- In contrast, dictionary methods rely on replacing substrings in a text with codewords, or indices, that identify the substring in a dictionary.
- Adaptive dictionary methods, like LZ77, are highly effective and rely on replacing substrings with pointers to earlier occurrences of the same substrings within the text being encoded.
- Symbolwise models may be improved by
  - moving from zero order models, which model probability of symbols without regards to context, to models, like ppm, which use higher order models that take left context into account;
  - using word-based models.

Summary (cont)

- To support random access into compressed files, techniques like synchrhonisation points or self-syncrhonising codes need to be used to avoid having to decompress an entire compressed archive to find a specific document or string.
- Performance of compression algorithms needs to be assessed by taking into account both the time taken to compress/decompress a file and the percentage reduction in filesize achieved.

Reading

- R. Baeza-Yates and B. Ribeiro-Neto (2011). Modern Information Retrieval: The Concepts and Technology Behind Search (ACM Press Books), 2nd edition. Chapter 6.8.
- I. H. Witten, A. Moffat, T. C. Bell. Managing Gigabytes: Compressing and Indexing Documents and Images, 2nd ed. Morgan Kaufmann. 1999.